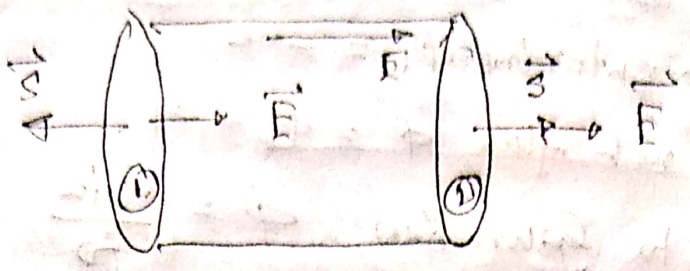
Electric flux. Gauss Law Solution

(3)



$\vec{S}$ = crossectional area of cylinder.

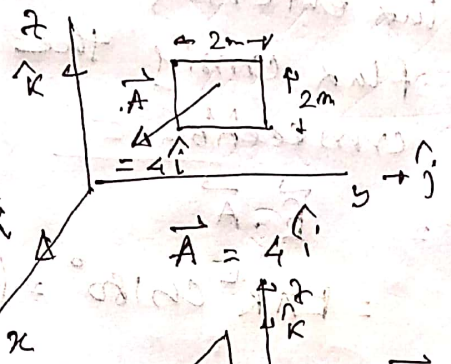
Electric field $\vec{E}$ or electric lines of forces are tangential to curved surface, So no flux through the curved surface.

$\therefore$ Net flux = flux entering + flux leaving

$= ES\cos 180^\circ + ES\cos 0^\circ$

$= -ES + ES = \underline{0}$

(1)



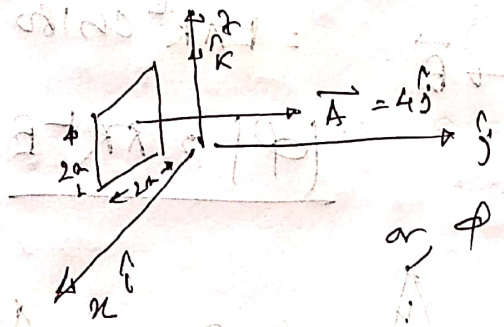
(i) $\phi = \vec{E}\cdot\vec{A}$

$\phi = (4\hat{j} + 3\hat{k})\cdot 4\hat{i}$

$= \underline{0}$

$\vec{A} = 4\hat{i}$

(ii)



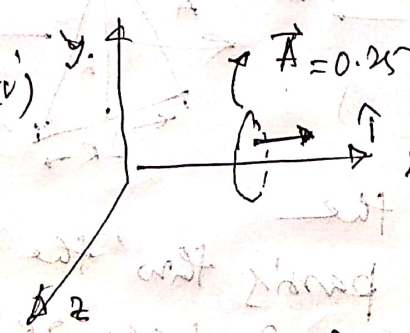
$\phi = \vec{E}\cdot\vec{A}$

or $\phi = (4\hat{j} + 3\hat{k})\cdot 4\hat{j}$

$= \underline{16 \text{ V-m}}$

(2)

(i)



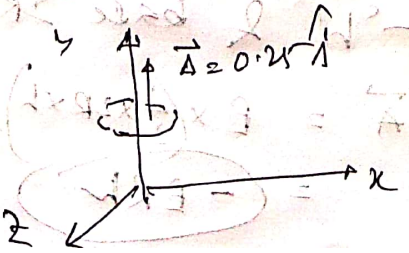
$\vec{E} = 20\hat{i} + 30\hat{j} - 40\hat{k}$

$\phi = \vec{E}\cdot\vec{A}$

$= (20\hat{i} + 30\hat{j} - 40\hat{k})\cdot 0.25\hat{i}$

$= \underline{5 \text{ V-m}}$

(ii)



$\phi = \vec{E}\cdot\vec{A}$

$= (20\hat{i} + 30\hat{j} - 40\hat{k})\cdot 0.25\hat{j}$

$= 30 \times \frac{1}{4} = 7.5 \text{ V-m}$

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(4) flux leaving ϕ_2 = +ve

flux entering ϕ_1 = −ve

So net flux $\phi = \phi_2 - \phi_1$

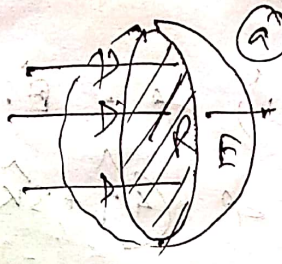
Acc to Gauss Law, $\phi = \frac{q_{in}}{\epsilon_0}$

$\therefore q_{in} = \epsilon_0 \phi$

$$\boxed{q_{in} = \epsilon_0 (\phi_2 - \phi_1)}$$

(5) net charge in a single dipole $+q - q = 0$, So for the dipoles $q_{inside} = 0$ $\therefore$ flux $= \frac{q_{in}}{\epsilon_0} = 0$

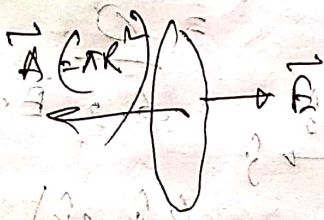
(6)



(a) Flux entering = flux crossing the cross section

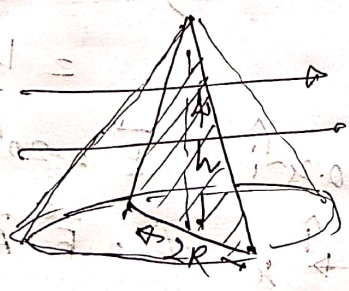
$\phi = \vec{E} \cdot \vec{A}$

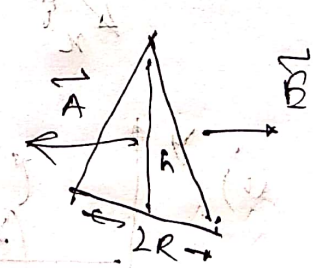
$= E \pi R^2 \cos 180° = -\pi R^2 E$



$$|\phi| = \pi R^2 E$$

(b)





Flux entering the cone = Flux passing thro' the △ triangle of ht 'h' & base '2R'

$\phi = \vec{E} \cdot \vec{A} = E \times \left(\frac{1}{2} \times 2R \times h\right) \cos 180°$

$= -ERh$

$\therefore |\phi| = ERh.$

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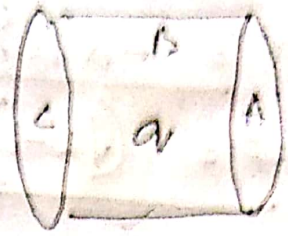
(7) [Flux] through C = [Flux] thro' A = ϕ_A say.



∴ By Gauss law

net flux $= \phi_C + \phi_A + \phi_B = \frac{q_{in}}{\epsilon_0}$

$[\because \phi_C = \phi_A]$ or, $2\phi_A + \phi = \frac{q}{\epsilon_0}$

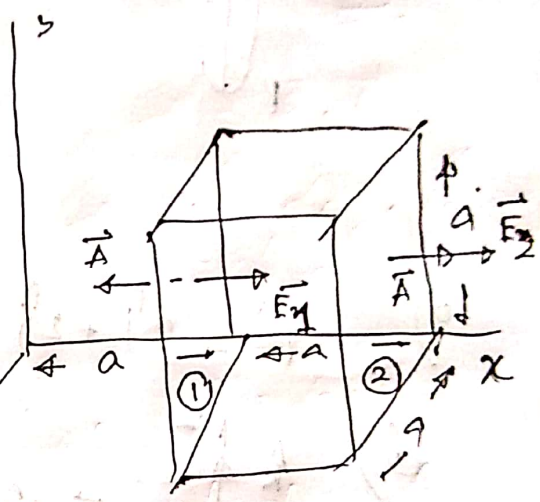
$\phi_B = \phi$

$$\boxed{\phi_A = \frac{1}{2}\left(\frac{q}{\epsilon_0} - \phi\right)}$$

(8) $|\vec{A}| = a^2$, $E_y = 0$, $E_z = 0$

$E_x = \alpha x^{1/2}$.

Apart from left face (1) & right face (2) of the cube, 'Ex' is tangential to other four faces, so no flux thro' them.



Electric field at left face (1) $E_1 = \alpha a^{1/2}$ ∵ at left face (1) $x = a$.

So flux entering left face (1) $\phi_1 = \vec{E_1} \cdot \vec{A}$

$\phi_1 = \alpha a^{1/2} \times a^2 \times \cos 180^\circ$

$= (-\alpha a^{5/2})$

Electric field at right face, (2) $E_2 = \alpha (2a)^{1/2}$ ∵ at right face (2) $x = 2a$.

So flux leaving the right face (2) $\phi_2 = \vec{E_2} \cdot \vec{A}$

$\phi_2 = \alpha (2a)^{1/2} \times a^2 \times \cos 0^\circ$

$= +\sqrt{2}\,\alpha a^{5/2}$

So net flux crossing cube $\phi = \phi_1 + \phi_2 =$

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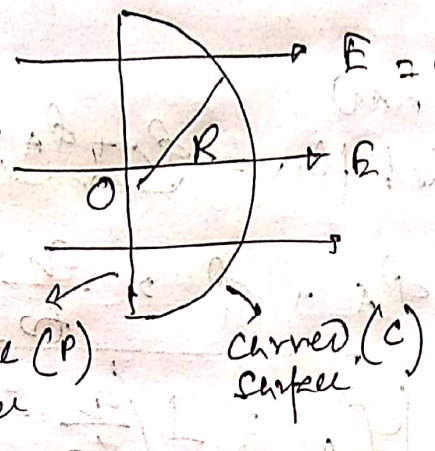
$$\phi = -\alpha a^{5/2} + \sqrt{2}\alpha a^{5/2}$$

$$\boxed{\phi = (\sqrt{2}-1)\alpha a^{5/2}}$$

(9)



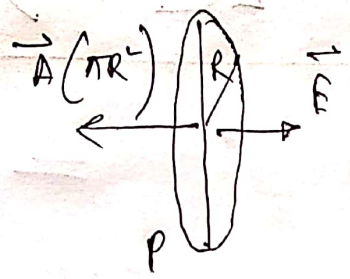
By Gauss's law,

$$\phi_{net} = \frac{q_{in}}{\epsilon_0}$$

$$\phi_p + \phi_c = \frac{q_{in}}{\epsilon_0}$$

$\because q_{inside} = 0$

$$\therefore \phi_c = -\phi_p$$



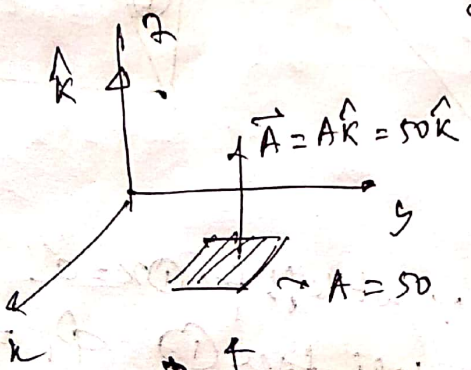
$$= -\left[\vec{E}\cdot\vec{A}\right]$$

$$= -\left[E \times \pi R^2 \times \cos 180^\circ\right]$$

$$= \underline{\pi R^2 E}$$

10) $\vec{E} = 3\hat{i} + 2\hat{j} + \hat{k}$

(i)

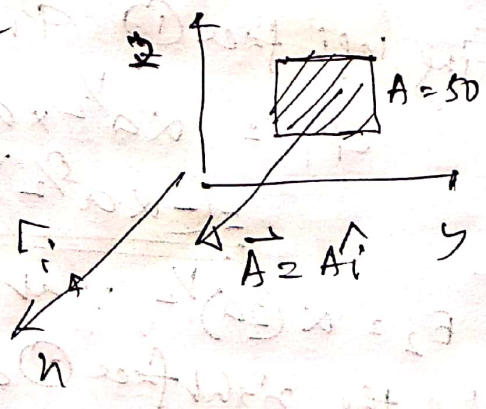


$$\phi = \vec{E}\cdot\vec{A}$$

$$= (3\hat{i} + 2\hat{j} + \hat{k})\cdot 50\hat{k}$$

$$= 50 \text{ V-m.}$$

(ii)



$$\phi = \vec{E}\cdot\vec{A}$$

$$= (3\hat{i} + 2\hat{j} + \hat{k})\cdot 50\hat{i}$$

$$= 150 \text{ V-m.}$$

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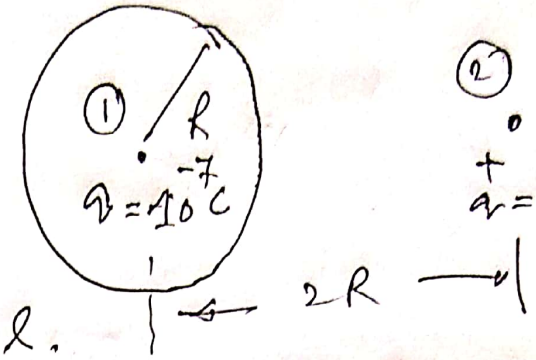
(11)

Contribution of electric flux thro' the sphere due to electric field of outer charge (2) is nil.

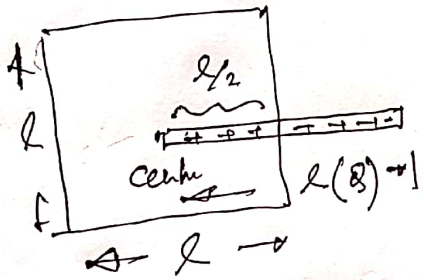


$\therefore$ net flux through sphere $= \dfrac{q_{inside}}{\epsilon_0}$

$= \dfrac{10^{-7}}{8.85 \times 10^{-12}}$

$= \underline{1.1 \times 10^{4}}$ N-m²/C.

(12)



charge/length of the rod $= Q/l$

$\therefore$ charge inside cube

$q_{in} = \left(\dfrac{Q}{l}\right) \times \dfrac{l}{2} = \dfrac{Q}{2}$

Here $\phi_{cube} = \dfrac{q_{in}}{\epsilon_0} = \boxed{\dfrac{Q}{2\epsilon_0}}$

(13) $\phi_{cube} = \dfrac{q}{\epsilon_0}$, $\phi_{face} = \dfrac{q}{6\epsilon_0}$.

(14)
(a) False
(b) True
(c) False
(d) True
(e) True

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